

Counting Past the Boundary

Enumerative CFAC for LERW in $d=3$, TAP Complexity, and KPZ Upper Tails

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Abstract

The CFAC stratification theorem (Amarteifio, 2026a) has a narrow domain: polynomial Dyson–Schwinger equations with algebraic or D -finite generating functions and a scalar rank- k bridge. The companion paper (Amarteifio, 2026b) marks three cases beyond that boundary — loop-erased random walk in three dimensions, TAP complexity in spin glasses, and the KPZ upper-tail rate function. Here we revisit those cases under the broader CFAC umbrella, where the theorem is one deliverable and *weighted enumeration of combinatorial objects* is another. For each case we extract a clean generating-function statement: (i) the LERW two-point function on any finite graph is a ratio of Kirchhoff polynomials, so the 3D exponent ν is a finite-size scaling question about a concrete rational function; (ii) the Kac–Rice expected complexity for p -spin models is (up to a Legendre transform) a ribbon-graph enumeration, evaluated via Harer–Zagier; (iii) the replicated lattice directed polymer is a chemical reaction network of n distinguishable walkers with pairwise δ -attraction, with Schur-process structure. None of these produces the open exponent in closed form, but each gives the object under study a concrete combinatorial definition that the CFAC pipeline can operate on at finite size. The purpose is to widen the scope statement honestly: the theorem does not reach these problems, but the enumeration layer does.

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1 Introduction: two tiers of CFAC

The CFAC programme, as stated in (Amarteifio, 2026a), has two distinct deliverables that are usually discussed together.

Tier A (stratification theorem). For a polynomial Dyson–Schwinger equation $G = z\phi(G)$ with an algebraic or D -finite generating function, the Puiseux order k of the dominant branch point gives a density-exponent skeleton $\tau_0 = 1 + 1/k$, with a one-loop dressing $\gamma_k(d)$ from a scalar rank- k bridge function.

Tier B (enumerative combinatorics). The underlying CFAC pipeline treats Feynman diagrams as weighted combinatorial objects — Symanzik polynomials, matroidal deletion–contraction, BPHZ multiplicities, sector decomposition — and computes exact finite-size generating functions independently of whether a closed-form asymptotic exists.

Background: unpacking the Tier-A statement

The Tier-A sentence compresses several technical notions that are worth spelling out for readers coming from physics or probability rather than analytic combinatorics.

Generating function (GF). A formal power series $F(z) = \sum_{n \geq 0} a_n z^n$ whose coefficients a_n count the combinatorial configurations of size n in the problem of interest — for us, Feynman diagrams with n loops, walks of length n , spanning trees of given weight, etc. The variable z has no physical meaning; it is a book-keeping marker. For a physicist, $F(z)$ plays the role of a partition function in a fictitious ensemble with fugacity z for “size”. The asymptotic behaviour of a_n as $n \rightarrow \infty$ — which is where physics lives — is encoded in the nearest singularity of $F(z)$ on the circle of convergence: location z_c sets the exponential growth $a_n \sim z_c^{-n}$, and the *type* of singularity sets the subleading power.

Algebraic GF. $F(z)$ is *algebraic* if it satisfies a polynomial equation $P(z, F(z)) = 0$ for some $P \in \mathbb{Q}[z, y]$. The simplest example is the Catalan-number GF $F = 1 + zF^2$, solution $F(z) = (1 - \sqrt{1 - 4z})/(2z)$, with a square-root branch point at $z_c = 1/4$. Algebraic means “closed-form in polynomials and n -th roots”. Many tree-like combinatorial classes have algebraic GFs; all of them admit exact asymptotic extraction by singularity analysis.

D -finite GF. $F(z)$ is *D -finite* (equivalently, *holonomic*) if it satisfies a linear ordinary differential equation with polynomial coefficients: $p_r(z)F^{(r)}(z) + \dots + p_0(z)F(z) = 0$ with $p_i \in \mathbb{Q}[z]$. Every algebraic GF is D -finite, but the converse fails: $\exp(z)$, Bessel functions, hypergeometric series, and solutions of most special-function ODEs are D -finite but not algebraic. For a physicist, D -finite is the class where the function is “solvable by a finite ODE with polynomial coefficients”, and its coefficients satisfy a finite linear recurrence with polynomial coefficients. *Not D -finite* means: no such finite ODE exists; the function lies outside the reach of classical special-function methods. Lattice Green’s functions on $\mathbb{Z}^d$ for $d \geq 2$ are the canonical non- D -finite examples in physics.

Dyson–Schwinger equation (DSE). In quantum field theory a DSE is a self-consistent integral equation relating a full Green’s function to itself, obtained by resumming an infinite tower of Feynman diagrams. In our combinatorial setting, “DSE” specialises to a functional equation of the form

$$G(z) = z\phi(G(z)),$$

where ϕ is a polynomial (or more general analytic function) of G . This is the “polynomial DSE” of Tier A. It encodes rooted tree-like combinatorial structures: the root contributes one z , each child subtree contributes a factor of G , and the vertex structure lives in ϕ . The Catalan recursion $G = z(1 + G^2)$ (rooted binary trees) and more generally any vertex polynomial give specific instances. Self-similar random-cluster models, directed-percolation mean-field recursions, and Doi–Peliti tree-level vertex resummations all produce DSEs of this shape.

Puiseux order k at the dominant singularity. Near its nearest singularity z_c an algebraic GF admits a *Puiseux expansion* — a power series in fractional powers of $(z_c - z)$:

$$G(z) = G(z_c) - A(z_c - z)^{1/k} + \dots, \quad k \in \mathbb{Z}_{\geq 2}.$$

The integer k is the *Puiseux order*. For $k = 2$ the leading behaviour is a square-root cusp, for $k = 3$ a cube-root cusp, and so on. By singularity analysis the coefficients behave as $a_n \sim Cz_c^{-n} n^{-1-1/k}$; the exponent $\tau_0 = 1 + 1/k$ appearing in Tier A is this. Different universality classes correspond to different k : mean-field directed percolation has $k = 2$, tricritical branching has $k = 3$, and so on. The Puiseux order k is the *CFAC*

stratification index: a discrete invariant separating the space of polynomial DSEs into strata, within each of which the leading exponent τ_0 is universal.

Scalar rank- k bridge. Beyond the leading exponent $\tau_0 = 1 + 1/k$, the CFAC programme refines the universal answer with a one-loop correction $\gamma_k(d)$ encoding the spatial dimension d . The object that carries this correction is a *bridge function* $B_k(r)$: a universal single-variable function of a dimensionless ratio r (mass or diffusion ratio, etc.) determined by the rank k and independent of the specific DSE. “Scalar” means B_k is number-valued (the bridge is a scalar function of one argument); “rank- k ” refers to the Puiseux order k that classified the DSE. The bridge converts the counting exponent τ_0 into a physical density exponent, with explicit dimensional dependence $\gamma_k(d)$ inserted at the one-loop order. The failure modes in the three boundary problems of the companion paper (Amarteifio, 2026b) are precisely failures of this scalar bridge to close: the problem demands a bridge of higher tensor rank (TAP), a disorder-averaged bridge (KPZ upper tail), or there is simply no DSE of the polynomial form at all (LERW in $d = 3$ — the GF is non- D -finite).

Stratification. Putting these together: the CFAC theorem *stratifies* the space of polynomial DSEs by the Puiseux order k of the dominant singularity, and attaches to each stratum a canonical pair (τ_0, γ_k) — the skeleton exponent and the one-loop dressing from the scalar rank- k bridge. The word “stratification” is used in the classical sense (decomposition of a space into lower-dimensional pieces indexed by a discrete invariant), not in any sheaf-theoretic sense.

The companion paper (Amarteifio, 2026b) flags three problems where Tier A does not reach:

- LERW in $d = 3$: the thermodynamic-limit generating function on $\mathbb{Z}^3$ is a lattice Green’s function, neither algebraic nor D -finite.
- TAP complexity: the organising quantity $\mathbb{E}|\det(H - fI)|$ needs a matrix-valued bridge; the current bridge is scalar.
- KPZ upper tail: the coefficient of the $|h|^{3/2}$ rate function requires a disorder-averaged bridge; current bridges are deterministic.

Each obstruction is a Tier-A obstruction. In this note we ask the Tier-B question for the same three problems: *what weighted enumeration does the generating-function language hand us?* The answers are known to specialists in each subfield; the contribution here is to show that they align under a single CFAC lens, giving concrete objects the existing machinery can compute at finite size.

Throughout, “CFAC enumeration” means the diagrammatic / combinatorial content of the pipeline — Symanzik polynomials, matrix-tree determinants, weighted path and partition sums — with no appeal to the stratification theorem.

2 The enumerative layer

Three constructions recur below. Fixing them here lets the three case sections stay short.

Kirchhoff / first Symanzik polynomial. For a finite graph $G = (V, E)$ with edge weights $w : E \rightarrow \mathbb{R}_{\geq 0}$, the *weighted spanning-tree polynomial* is

$$\tau(G; w) = \sum_{T \in \mathcal{T}(G)} \prod_{e \in T} w_e, \quad (1)$$

where $\mathcal{T}(G)$ is the set of spanning trees. The *matrix-tree theorem* gives a closed form: $\tau(G; w) = \det L^{(v)}(G; w)$ for any reduced weighted Laplacian $L^{(v)}$. In parametric Feynman integration, Eq. (1) is the first Symanzik polynomial; in CFAC it is the object produced by `rdft/ac/dse.py` after matroidal deletion-contraction.

Map / ribbon-graph enumeration. For an $N \times N$ symmetric Gaussian random matrix M (GOE),

$$\mathbb{E}[\text{tr } M^{2k}] = \sum_{g \geq 0} N^{k+1-2g} \varepsilon_g(k), \quad (2)$$

where $\varepsilon_g(k)$ is the number of rooted one-face maps with k edges on a (possibly non-orientable) surface of genus g . The orientable-surface case is the Harer–Zagier formula (Harer and Zagier, 1986); the non-orientable extension is due to (Goulden and Jackson, 1997). In CFAC these are the Feynman diagrams of the matrix-model action $\frac{1}{2} \text{tr } M^2 - \frac{g}{N} \text{tr } M^k$, computed as ordinary (weighted) graph counts.

Weighted path enumeration with δ -interactions.

Fix a transition kernel $p(x, y)$ on a finite state space. The partition function of n distinguishable walkers of length N with a pairwise attractive contact interaction of strength β^2 is

$$Z_N^{(n)}(\beta) = \sum_{\omega^1, \dots, \omega^n} \prod_{t=1}^N p(\omega_{t-1}^i, \omega_t^i) e^{(\beta^2/2) k_t(\omega)}, \quad (3)$$

where $k_t(\omega) = \sum_x (\#\{i : \omega_t^i = x\})^2$ is the coincidence count at time t . The reaction-network reading is a CRN with n species and a symmetric binary vertex $i + j \rightarrow i + j$ carrying rate β^2 . Doi–Peliti quantisation of this CRN gives the replica moment of a directed polymer (g5).

Each case below reduces to one of these three constructions.

3 LERW in $d = 3$ as Kirchhoff enumeration

Setup. Let $G = (V, E)$ be a finite connected weighted graph and fix two vertices $a, b \in V$. The loop-erased random walk $\text{LERW}_{a \rightarrow b}$ is the law of the chronologically loop-erased simple random walk from a conditioned to hit b . By Wilson’s algorithm (Wilson, 1996),

$\text{LERW}_{a \rightarrow b}$ coincides with the law of the unique a -to- b path in the uniform spanning tree of G .

Proposition 1 (LERW as spanning-tree ratio). *For any simple path γ from a to b in G ,*

$$\mathbb{P}[\text{LERW}_{a \rightarrow b} = \gamma] = \frac{w(\gamma) \tau(G/\gamma)}{\tau(G)}, \quad (4)$$

where $w(\gamma) = \prod_{e \in \gamma} w_e$ and G/γ is the contraction of G along γ .

Proof sketch. The event $\{\text{LERW}_{a \rightarrow b} = \gamma\}$ is the event that the unique a - b path in the UST equals γ , which is the event $\{\gamma \subset \text{UST}\}$. The conditional law of the UST given $\gamma \subset \text{UST}$ is uniform over spanning trees of G/γ . The unconditional UST weight of a fixed spanning tree T is $w(T)/\tau(G)$, so $\mathbb{P}[\gamma \subset \text{UST}] = w(\gamma)\tau(G/\gamma)/\tau(G)$. $\square$

Length generating function. Summing Eq. (4) against a length marker gives a rational function in z :

$$F_{ab}(z; G) \equiv \sum_{\gamma: a \rightarrow b \text{ simple}} z^{|\gamma|} w(\gamma) \frac{\tau(G/\gamma)}{\tau(G)}. \quad (5)$$

Corollary 2 (Rational representation on finite graphs). *For any finite G , $F_{ab}(z; G)$ is a rational function in z with denominator $\tau(G)$. Both numerator and denominator are evaluations of the weighted Laplacian determinant.*

The corollary is just bookkeeping on Eq. (4), but the content is that the LERW two-point function is a quantity CFAC already computes: the ratio of two first Symanzik polynomials.

Scaling limit as finite-size scaling of F_{0b} . Take $G = \mathbb{Z}_N^3$ (the N^3 -torus). Let $\langle |\gamma| \rangle_{0b}$ denote the LERW expected length from 0 to b under Eq. (4). The 3D LERW scaling exponent (Kozma, 2007) can be read off from

$$\langle |\gamma| \rangle_{0b} \asymp |b|^{1/\nu_3}, \quad \nu_3 \approx 1.624, \quad (6)$$

which by Corollary 2 is a finite-size scaling statement about $z \partial_z \log F_{0b}(z; \mathbb{Z}_N^3) \big|_{z=1}$ as $N, |b| \rightarrow \infty$ in the appropriate order. The obstruction the stratification theorem meets — that the thermodynamic limit is a non- D -finite lattice Green’s function — does not obstruct this finite-size object, which is a ratio of explicit determinants.

Tropical / monomial-dominance limit. In the limit where one spanning tree dominates (e.g. when edge weights are $e^{-\beta E_e}$ and $\beta \rightarrow \infty$), Eq. (4) reduces to a deterministic path: the unique minimum-weight spanning tree. The LERW distribution collapses onto the tree-path in this limit. This is the tropical Kirchhoff limit mentioned in (Amarteifio, 2026b, §7.1); it is a well-defined CFAC enumeration in its own right.

Remark 1. Prop. 1 is classical (it is Wilson’s algorithm in reverse). The CFAC reading is that Eq. (4) places LERW squarely inside the first-Symanzik-polynomial enumeration tier: the same machinery that produces two-loop hat diagrams (Amarteifio, 2026c) produces the LERW two-point function, with no changes beyond the graph on which it is evaluated.

Why $d = 3$ is the hard case: dimensional partitioning of the LERW exponent. The exponent ν_d is tractable in every dimension *except* the physically relevant one, and the reason is structural:

- **$d = 2$: conformal symmetry.** The LERW scaling limit is SLE_2 (Lawler, Schramm and Werner, 2004), an instance of the conformally-invariant family SLE_κ . Conformal invariance in two dimensions is generated by an infinite-dimensional Lie algebra (Virasoro), enough symmetry to collapse every observable to a formula in the single parameter κ . For LERW this gives $d_f = 1 + \kappa/8 = 5/4$ in closed form, and with it $\nu_2 = 4/5$.
- **$d \geq 4$: mean-field / dimensional decoupling.** Above the upper critical dimension $d_c = 4$, two independent simple random walks typically do not intersect, so loop-erasure acts trivially in the bulk. LERW behaves as SRW with at most logarithmic corrections, and $d_f = 2$ is forced by the Gaussian scaling of SRW (Lawler, 1986). At $d = 4$ this is exact up to $\log^{1/3}$ corrections; for $d \geq 5$ it is exact.
- **$d = 3$: no symmetry, no small parameter.** In three dimensions the conformal group reduces to a finite-dimensional Möbius group (insufficient to fix any exponent), and the interaction is not small (two 3D random walks intersect with positive probability). Kozma’s theorem (Kozma, 2007) proves the scaling limit exists and is supported on a set of some Hausdorff dimension $d_f^{(3)}$, numerically ≈ 1.6236 (Wilson, 2010), but the proof is non-constructive — no technique is known that produces $d_f^{(3)}$ in any form, closed or otherwise. The same obstruction is responsible for the unresolved status of 3D Ising, 3D percolation, and 3D Heisenberg critical exponents.

The obstruction is therefore *dimension-specific* and *tool-specific*: 2D has conformal algebra, $d \geq 4$ has dimensional decoupling, 3D has neither. The finite-graph Kirchhoff ratio of Prop. 1 is exact in every dimension; the scaling-limit extraction loses closed-form tractability only at $d = 3$. Section ?? returns to this from the analytic-combinatorics angle: given that the finite-size generating function is known exactly, what does asymptotic extraction from it give us, even without a closed form for the limit?

4 TAP complexity as ribbon-graph enumeration

Setup. For the spherical pure p -spin Hamiltonian H_N on $S^{N-1}(\sqrt{N})$, the Kac–Rice formula for the expected number of critical values in $[NE_1, NE_2]$ is

$$\mathbb{E}N_{\text{crit}}([E_1, E_2]) = c_{N,p} \int_{E_1}^{E_2} \mathbb{E}|\det(M - f(E)I)| \rho_E(E) dE, \quad (7)$$

where M is a $\text{GOE}(N-1)$ matrix, $f(E)$ is a linear shift fixed by the Lagrange multiplier for the spherical constraint, and $c_{N,p}, \rho_E$ are explicit Gaussian factors (Auffinger, Ben Arous and Černý, 2013; Subag, 2017).

Characteristic-polynomial expectation as a generating function. For $\text{GOE}(N)$,

$$\log \mathbb{E}|\det(M - fI)| = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k f^k} \mathbb{E} \text{tr}[(M/\sqrt{N})^k] + O(1/f), \quad (8)$$

convergent for $|f|$ above the semicircle edge. The coefficients $\mathbb{E} \text{tr}[(M/\sqrt{N})^k]$ admit exact finite- N expansions into rooted map counts (Harer and Zagier, 1986; Goulden and Jackson, 1997):

$$\mathbb{E}[\text{tr}(M/\sqrt{N})^{2k}] = \sum_{g \geq 0} N^{-2g} \varepsilon_g^{\text{GOE}}(k), \quad (9)$$

where $\varepsilon_g^{\text{GOE}}(k)$ counts rooted one-face maps with k edges on a (possibly non-orientable) surface of genus g .

Proposition 3 (TAP complexity as map enumeration). *The Kac–Rice complexity integrand in Eq. (7) is the exponential of a generating function over rooted one-face maps:*

$$\log \mathbb{E}|\det(M - fI)| = N \Phi_0(f) + \Phi_1(f) + N^{-2} \Phi_2(f) + \dots, \quad (10)$$

with each Φ_g determined by the map-counting coefficients $\{\varepsilon_g^{\text{GOE}}(k)\}_{k \geq 1}$ via Eq. (8) and the moment expansion Eq. (9). At large N , $\Phi_0(f) = \int \log |\lambda - f| \rho_{\text{sc}}(\lambda) d\lambda$, recovering the semicircle complexity density of (Auffinger, Ben Arous and Černý, 2013).

Proof. Cluster expansion of $\log \mathbb{E} \exp[\sum_k \lambda_k \text{tr} M^k]$ against the genus expansion of (Harer and Zagier, 1986; Goulden and Jackson, 1997); specialise λ_k to the coefficients of $\log |\det(M - fI)|$ from Eq. (8). The large- N limit of Φ_0 is the Wigner semicircle transform because the planar ($g = 0$) map count is the Catalan numbers, whose generating function is $(1 - \sqrt{1 - 4t})/(2t)$. $\square$

CFAC hookup. Φ_0 is the standard TAP complexity density. Prop. 3 places it in the CFAC enumeration tier: the finite- N and subleading corrections are exact sums over rooted one-face maps, which are diagrams the pipeline handles natively (they are the Feynman diagrams of a matrix ϕ^3/ϕ^4 model). The obstruction flagged in (Amarteifio, 2026b, §7.2) — that a matrix-valued bridge is needed to close the scaling of $\Sigma(E)$

via the stratification theorem — does not apply at this tier: one is counting maps, not extracting a Puiseux exponent.

Remark 2. The Legendre transform $\Sigma(E) = \sup_f [-R(E, f) + \Phi_0(f)]$, with $R(E, f)$ the Gaussian rate (Auffinger, Ben Arous and Černý, 2013), is the large-deviation form of TAP complexity. In the map-counting language, this is the saddle point of a generating function over rooted ribbon graphs with an external parameter f tuned to the complexity curve. The complexity threshold E_∞ (below which critical points cease to be exponentially numerous) is the value of E at which the saddle hits the edge of the semicircle, i.e. the radius of convergence of the planar map generating function.

5 KPZ upper tail as replica CRN enumeration

Setup. The (1+1)-dimensional lattice directed polymer with i.i.d. Gaussian weights $\eta(t, x)$ of variance 1 has partition function

$$Z_N(\beta) = \sum_{\omega: (0,0) \rightarrow (N, \cdot)} \prod_{t=0}^N \exp[\beta \eta(t, \omega_t)], \quad (11)$$

with sum over nearest-neighbour lattice paths. The KPZ universality hypothesis predicts that for large N , $\log Z_N - c_1 N \sim c_2 N^{1/3} \chi$ with χ Tracy–Widom-distributed, and an upper-tail rate function $\mathbb{P}[\log Z_N - c_1 N > y N^{1/3}] \asymp e^{-c_3 y^{3/2}}$ at large positive y (Corwin and Ghosal, 2020; Krajenbrink, 2020).

Replicated moments as weighted enumeration. For integer $n \geq 1$, disorder-averaging gives

$$\mathbb{E}[Z_N(\beta)^n] = \sum_{\omega^1, \dots, \omega^n} \prod_{t=0}^N \exp\left[\frac{\beta^2}{2} k_t(\omega^{1:n})\right], \quad (12)$$

with the coincidence count $k_t(\omega^{1:n}) = \sum_x (\#\{i : \omega_t^i = x\})^2$. Eq. (12) is an instance of the weighted n -walker enumeration Eq. (3).

Proposition 4 (KPZ replica moments as CRN partition function). *For each integer $n \geq 1$, $\mathbb{E}[Z_N^n]$ is the partition function of a chemical reaction network with n distinguishable random-walker species and a symmetric binary contact vertex $i + j \rightarrow i + j$ of rate β^2 , on N time steps. The Doi–Peliti representation is an MSR action with kinetic term for each walker and a single δ -localised four-leg vertex identifying species pairwise.*

The proof is direct inspection of Eq. (12). In the continuum limit the CRN becomes the attractive n -body Lieb–Liniger model, and this is the classical replica route to Tracy–Widom (Krajenbrink, 2020).

Schur-process form (discrete geometries). For several exactly-solvable lattice models (log-gamma polymer, O’Connell–Yor, geometric last passage), RSK

correspondence converts the replica sum into a partition sum (Borodin and Corwin, 2014):

$$\mathbb{E}[Z_N^n] = \sum_{\lambda \vdash n} s_\lambda(x_1, \dots, x_N) s_\lambda(y_1, \dots, y_N) \cdot (\text{spectral weight}), \quad (13)$$

with s_λ the Schur polynomial. Eq. (13) is an enumeration over partitions weighted by Schur pairs, and the generating function $\sum_n \lambda^n / n! \mathbb{E}[Z_N^n]$ becomes a Schur measure on partitions — a determinantal point process whose right edge, under an appropriate scaling, is the Airy_2 kernel underlying Tracy–Widom.

Upper-tail exponent as a Schur-measure edge statement. The $|y|^{3/2}$ upper-tail shape arises as the large-deviation form of the right edge of the Schur / Plancherel measure on partitions, an instance of the edge large-deviation theory of determinantal processes. This is where the *coefficient* problem sits: translating between lattice weight distributions and Schur specialisations.

Proposition 5 (Exponent vs. coefficient). *At the exponent level, the $|y|^{3/2}$ upper-tail shape follows from any discretisation admitting the Schur-process resummation Eq. (13) together with Airy-kernel edge asymptotics of the corresponding Schur measure. The model-specific coefficient c_3 depends on the spectral weight and is not determined by the enumeration alone.*

Proof sketch. Given Eq. (13), the generating function $\mathcal{G}(\lambda) = \sum_n \lambda^n \mathbb{E}[Z_N^n] / n!$ is a Fredholm determinant for a Schur kernel (Borodin and Corwin, 2014). The upper-tail rate of $\log Z_N$ corresponds to the rate of decay of $\mathbb{P}[\text{top row of } \lambda > L]$ under the Schur measure. The classical edge large-deviation result for such measures gives a rate function cubic in the scaled variable, yielding the $|y|^{3/2}$ shape under the appropriate exponentiation (Krajenbrink, 2020). The coefficient depends on the derivative of the spectral density at the edge, which is a model input. $\square$

CFAC hookup. The CRN of Prop. 4 sits inside the CFAC pipeline without modification: it is a Doi–Peliti action with n species and one vertex. The enumeration of configurations at fixed n is a standard walker counting. What is *not* inside the current pipeline is the replica continuation $n \rightarrow 0$ and the disorder-averaged bridge that would fix c_3 from first principles.

Tier-A closed form for the 2-body binding on $\mathbb{Z}$. At $n = 2$ the CRN reduces to a tight-binding chain with an attractive on-site impurity, which admits a closed-form bound-state eigenvalue. This is the one place where the enumeration layer delivers an exact Tier-A statement for the boundary cases.

Theorem 6 (Closed-form 2-body replica rate). *For the lattice directed polymer on $\mathbb{Z}$ with Gaussian disorder of variance 1 (coincidence weight $\exp(\frac{\beta^2}{2} k_t)$, $k_t = \sum_x (\#\{i : \omega_t^i = x\})^2$), the 2-body replica rate satisfies*

$$\begin{aligned} D(2, \beta; \mathbb{Z}) &\equiv \lambda(2, \beta; \mathbb{Z}) - 2\lambda(1, \beta; \mathbb{Z}) \\ &= 2\beta^2 - \log(2e^{\beta^2} - 1). \end{aligned} \quad (14)$$

The bound-state wavefunction in the relative coordinate $r = x_1 - x_2$ is $\psi(r) = z^{|r|/2}$ with $z = 1/(2e^{\beta^2} - 1)$.

Proof. Separate the translation-invariant centre-of-mass coordinate $s = (x_1 + x_2)/2$ from the relative coordinate r . At $k_s = 0$ the relative transfer matrix acts as $(T_{\text{rel}}\psi)(r) = \psi(r-2) + 2\psi(r) + \psi(r+2)$ for $r \neq 0$, with a multiplicative e^{β^2} at $r = 0$ from the coincidence weight. Ansatz $\psi(r) = z^{|r|/2}$ with $|z| < 1$ gives the bulk eigenvalue $\mu_{\text{rel}} = (1+z)^2/z$ for $r \neq 0$. The boundary condition at $r = 0$, $e^{\beta^2}(2z+2) = \mu_{\text{rel}}$, yields $2e^{\beta^2}(1+z) = (1+z)^2/z$, and hence $z = 1/(2e^{\beta^2} - 1)$. Substituting, $\mu_{\text{rel}} = 4e^{2\beta^2}/(2e^{\beta^2} - 1)$. Since $\lambda(1, \beta; \mathbb{Z}) = \beta^2/2 + \log 2$ and $\lambda(2, \beta; \mathbb{Z}) = \beta^2 + \log 2 + \log \mu_{\text{rel}} - \log e^{\beta^2}$ (the factored-out e^{β^2} per step and the COM factor $\log 2$ for $k_s = 0$), a direct computation gives Eq. (14). $\square$

Corollary 7 (Small- β asymptotics). As $\beta \rightarrow 0$,

$$D(2, \beta; \mathbb{Z}) = \beta^4 - \frac{\beta^6}{3} + \frac{\beta^8}{6} + O(\beta^{10}),$$

so the leading coefficient is β^4 , matching the Kardar / Bethe-ansatz scaling of the 2-body bound-state binding for 1D attractive δ -Bose gases. The bound state has localisation length $1/\log(2e^{\beta^2} - 1)$, diverging as $1/\beta^2$ at weak coupling and shrinking to $O(1)$ lattice sites at strong coupling.

Thm. 6 is verified against the numerical transfer-matrix spectrum: $D(2, \beta; W)$ is monotone decreasing in W and converges from above to Eq. (14) as $W \rightarrow \infty$. At $\beta = 0.8$ the closed form gives $D(2, 0.8; \mathbb{Z}) = 0.25290$, with the finite- W sequence 0.337, 0.310, 0.294, 0.284, 0.277, 0.272, 0.269 at $W = 3, \dots, 9$. Tests in `tests/test_replica_closed.py` verify the closed form, its series expansion coefficient by coefficient, and the finite- W convergence.

Status of the other boundary closed forms. The two remaining boundary exponents sit on opposite ends of the tractability spectrum, and it is worth being explicit about which is which.

KPZ upper-tail c_3 is closed in continuum. The coefficient $c_3 = 4/3$ is known exactly for the continuum KPZ equation (Das and Tsai, 2021; Krajenbrink and Le Doussal, 2021; Tsai, 2022), derived in multiple independent ways: each route goes through the Bethe-ansatz cubic-in- n shape $\lambda(n) = \frac{\beta^2}{2}n + \frac{\beta^4}{48}n(n^2 - 1)$, followed by a saddle-point / Airy-kernel edge argument on the resulting Fredholm determinant. The $3/2$ exponent is the Legendre conjugate of the cubic. For *lattice* polymers the value of c_3 is model-dependent; Thm. 6 already captures the $n = 2$ binding on $\mathbb{Z}$ in closed form, and the analogous three-body coordinate Bethe ansatz on $\mathbb{Z}$ (nested, tractable for on-site attraction) would close the $n = 3$ sector and give the lattice c_3 by the same Legendre route. This is routine extension of the machinery already in place rather than a conceptual obstacle.

LERW ν_3 is genuinely open. In $d = 2$ the exponent $\nu_2 = 4/5$ follows from the conformal-invariance

/ SLE₂ scaling limit (Lawler, Schramm and Werner, 2004), with the universal formula $d_f = 1 + \kappa/8$. In $d \geq 4$ the exponent $d_f = 2$ comes from the mean-field regime above the upper critical dimension (Lawler, 1986). In $d = 3$ neither mechanism applies: there is no conformal invariance in the scaling limit and no small parameter controlling the interaction. Kozma’s 2007 scaling-limit theorem (Kozma, 2007) is an *existence* result — it shows a limit exists and is supported on a set of some Hausdorff dimension — without producing ν_3 in any form. No closed form has been proposed even conjecturally. The same structural obstruction (below the upper critical dimension, no conformal symmetry in odd dimension) is what keeps the 3D Ising critical exponents open. For our programme this means: the finite-size Kirchhoff ratio of Prop. 1 is exact on every finite graph — the enumeration layer is complete — but the scaling limit on $\mathbb{Z}^3$ is beyond what any current technique can close.

6 Machine verification

The three propositions are backed by a passing pytest suite `tests/test_enumerative_boundary.py`, exercising three new modules:

- `rdft/ac/lerw.py` implements the Kirchhoff ratio of Prop. 1 and the length generating function Eq. (5). Tests verify $\tau(K_4) = 16$ (Cayley) and $\tau(C_n) = n$; that LERW probabilities sum to one on K_4 for several endpoint pairs; that the two arcs of C_6 from 0 to 3 carry equal mass $\frac{1}{2}$, while the short and long arcs from 0 to 1 carry unequal mass summing to one.
- `rdft/ac/matrix_model.py` implements the Harer–Zagier recursion for $\varepsilon_g(k)$ and the genus expansion of Eq. (9) (orientable / GUE case). Tests verify $\varepsilon_0(k) = C_k$ (Catalan), the classical values $\varepsilon_1(2, 3, 4, 5) = (1, 10, 70, 420)$, the identity $\sum_g \varepsilon_g(k) = (2k-1)!!$ at $N = 1$, and Wigner convergence $\mathbb{E} \text{tr} M^{2k}/N^{k+1} \rightarrow C_k$ as $N \rightarrow \infty$.
- `rdft/ac/replica.py` implements direct enumeration of $\mathbb{E}[Z_N^n]$ from Eq. (12) and a Monte Carlo cross-check by disorder averaging. Tests verify the $n = 1$ enumeration matches the transfer-matrix path count at $\beta = 0$, and that exact enumeration agrees with Monte Carlo within 3σ at $n = 1$ and $n = 2$ for $N = 3$, $\beta = 0.5$.

These are sanity checks for the combinatorial identities, not predictions of the open exponents.

7 Expressivity experiments

A second pass of modules pushes each case past the baseline check of §6 into a non-trivial numerical experiment.

LERW sampling on $\mathbb{Z}^d$ tori (`rdft/ac/lerw_scaling.py`). Lawler’s loop-erased random-walk construction samples the measure of

Prop. 1 directly: run SRW from a until it hits b , apply chronological loop erasure, record the path length. We verify on K_4 that sampled mean length agrees with the exact Kirchhoff ratio within one SEM, then sweep $\langle |\gamma| \rangle_{0 \rightarrow (r,0)}$ on $\mathbb{Z}_{16}^2$ and $\mathbb{Z}_8^3$. The mean length is strictly increasing in r in both dimensions. On a periodic torus (no killing boundary) the Kenyon slope $d_f = 5/4$ is not recovered at these finite sizes; the natural setting for Kenyon’s theorem is a box with Dirichlet boundary, handled next.

Dirichlet-box LERW and recovery of Kenyon / Kozma exponents (rdft/ac/lerw_dirichlet.py). For the killed-walk LERW on $[0, L-1]^d$ — start at the centre, run SRW with chronological loop erasure, stop the first time the walker steps outside the box — the mean path length scales as $\langle |\gamma| \rangle \asymp L^{d_f}$ with the lattice LERW fractal dimension (Kenyon, 2000; Kozma, 2007). A Monte-Carlo sweep at modest compute (1500–3000 samples per L) gives:

d	L range	Fit d_f	Ref. d_f
2	8–48	1.28	5/4
3	4–12	1.67	≈ 1.624
4	3–6	1.77	2 (mean field)

References: $d = 2$ (Kenyon, 2000); $d = 3$ (Kozma, 2007; Wilson, 2010).

The $d = 2$ slope sits 2.4% above the exact 5/4 at these box sizes, the $d = 3$ slope sits 2.8% above Kozma’s value, and the $d = 4$ trend toward the mean-field 2 is visible but finite-size dominated. This is the clean version of the numerical face of Prop. 1: the Kirchhoff-ratio measure, sampled on the graph where Kenyon’s theorem applies, recovers both known exponents within percent-level agreement from the CFAC enumeration layer alone.

Replica transfer matrix (rdft/ac/replica_transfer.py). The n -walker CRN of Prop. 4 is a transfer matrix on configuration space $\{-W, \dots, W\}^n$ of size $(2W + 1)^n$. The Perron eigenvalue gives the replica rate

$$\lambda(n, \beta; W) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \mathbb{E}[Z_N^n]. \quad (15)$$

Tests verify the free baseline $\lambda(n, 0; W) = n \log(2 \cos(\pi/(2W + 2)))$ (open-chain tight-binding), monotonicity in β , superlinear scaling $\lambda(n, \beta; W)/n$ increasing in n for $\beta > 0$ (replica attraction / binding), and exact agreement between the transfer-matrix partition function and the direct CRN enumeration of §6 for small (n, N) .

Cubic-in- n Bethe-ansatz signature (rdft/ac/replica_cubic.py). Kardar’s exact Bethe-ansatz solution of the continuum n -body attractive δ -Bose gas (Kardar, 1987; Calabrese, Le Doussal and Rosso, 2010) gives

$$\lambda(n, \beta) = a(\beta)n + b(\beta) \frac{n(n^2 - 1)}{6} + O(n^5), \quad (16)$$

the Lieb–Liniger ground-state shape that underlies the KPZ free-energy cumulants. Define $D(n) := \lambda(n, \beta; W) - n \lambda(1, \beta; W)$. The pure cubic ansatz predicts $D(2) = b$ and $D(3) = 4b$, so the model-independent witness $R(\beta, W) := D(3)/D(2) \rightarrow 4$ in the continuum limit $W \rightarrow \infty$. Exact transfer-matrix evaluation yields:

W	$D(2)$	$D(3)$	$R = D(3)/D(2)$	$ R - 4 $
3	0.337	1.100	3.261	0.739
4	0.310	1.040	3.352	0.648
5	0.294	1.006	3.420	0.580
6	0.284	0.985	3.470	0.530
7	0.277	0.972	3.507	0.493
8	0.272	0.962	3.535	0.465
9	0.269	0.956	3.556	0.444

(all at $\beta = 0.8$, exact eigenvalues). The ratio approaches 4 monotonically with the lattice width, with the finite- W correction empirically consistent with $|R - 4| \sim 1/W$ in the regime tested. Tests verify monotonicity in W , positivity of binding for $\beta > 0$, vanishing at $\beta = 0$, and that the lattice ratio sits strictly in $(3, 4)$ at every W in the tested range. This is the cleanest analytic handle on $\lambda(n, \beta; W)$ available from the transfer matrix at small n alone — the higher- n continuation that would close the upper-tail coefficient is the next analytic step.

Planar semicircle resolvent and Catalan reconstruction (rdft/ac/tap_complexity.py). The map-count generating function of Prop. 3 is tested at the planar order: the closed form of

$$I(f) = \int_{-2}^2 \log |\lambda - f| \rho_{\text{sc}}(\lambda) d\lambda \quad (17)$$

is matched against two independent computations. (i) Direct numerical integration of the defining integral, at a range of probe values $f \in \{-3, -2.5, -1.5, 0, 1.2, 2.5, 4\}$; agreement within 2×10^{-3} . (ii) Series reconstruction from Catalan numbers via

$$I(f) = \log |f| - \sum_{k \geq 1} \frac{C_k}{2k} f^{-2k}, \quad |f| > 2,$$

matched within 10^{-4} at $|f| \in \{2.5, 3, 4, 6\}$ with 80 Catalan terms. This is the genus-0 face of Prop. 3: Catalan counts (planar rooted one-face maps) reconstruct the semicircle resolvent.

Combinatorial-animal experiment for $d = 3$ LERW (rdft/ac/lerw_hierarchical.py). Since the $\mathbb{Z}^3$ thermodynamic generating function is not D -finite, standard analytic-combinatorics extraction does not terminate. The alternative route is to build a *combinatorial animal*: a self-similar graph sequence $\{G_k\}$ on which LERW is tractable, and measure the gap between the animal’s scaling exponent and the $\mathbb{Z}^3$ Monte-Carlo value $d_f \approx 1.624$. Two animals are implemented:

Migdal–Kadanoff diamond ($b = 4, s = 2$).

G_{k+1} replaces each edge of G_k by b parallel branches of s edges each. Effective dimension $d_{\text{eff}} = 1 + \log b / \log s = 3$. Every G_k is a series-parallel graph, so its UST deterministically selects one branch per diamond at every level and the source-to-target path realises exactly the graph diameter s^k . Hence $d_f^{\text{MK}}(4, 2) = 1$ exactly, a large-gap *negative result*: the MK approximant, despite having the correct $d_{\text{eff}} = 3$, loses the meandering structure of LERW entirely. The failure is structural: LERW is degenerate on every series-parallel graph.

Sierpiński gasket. G_{k+1} is the “triforce” arrangement of three copies of G_k glued at top-level corners. Hausdorff dimension $\log 3 / \log 2 \approx 1.585$. The gasket is not series-parallel: every sub-triangle carries a cycle, so loop erasure is active. Monte-Carlo extraction at levels $k = 2, \dots, 5$ gives $d_f^{\text{Sierp}} \approx 1.20$, strictly between 1 and the gasket’s Hausdorff dimension. The animal now has nontrivial loop-erasure exponent but at the wrong fractal dimension; its gap to $\mathbb{Z}^3$ ’s value 1.624 is ≈ 0.42 .

Animal	d_{eff}	d_f^{LERW}	Gap
MK diamond (4, 2)	3	1.00	−0.62
Sierpiński gasket	$\log_2 3$	≈ 1.20	−0.42
$\mathbb{Z}_L^3$ box	3	≈ 1.67	+0.05

The two animals break in qualitatively different ways. The MK diamond reproduces the *dimension label* of $\mathbb{Z}^3$ but not its graph structure (every loop closes trivially). The Sierpiński gasket reproduces the *loop-erasure mechanism* but at the wrong fractal dimension. $\mathbb{Z}^3$ demands both non-trivial loop structure and translation-invariant three-dimensional connectivity at once — neither animal captures both, which is an honest statement of what the obstruction to a closed-form $\mathbb{Z}^3$ exponent actually is.

Algebraic / compositional reading. Each hierarchical construction $G_{k+1} = \Phi(G_k)$ is a finite *compositional* rewrite in the sense of Flajolet–Sedgewick’s symbolic method: at the level of generating functions, the LERW length series $F_k(z)$ satisfies a functional recursion

$$F_{k+1}(z) = \Phi^\sharp[F_k(z)], \quad (18)$$

where $\Phi^\sharp$ is the rational transformation on generating functions induced by the graph rewrite Φ through the Kirchhoff-ratio definition of Prop. 1. The fractal exponent d_f is then the linearisation eigenvalue of $\Phi^\sharp$ at its renormalisation fixed point:

$$d_f = \frac{\log \lambda^\sharp}{\log s},$$

with $\lambda^\sharp$ the dominant eigenvalue of $d\Phi^\sharp$ and s the length-scale factor of Φ . This is the Kadanoff block-spin renormalisation rephrased as a fixed-point problem in the Flajolet symbolic category: the *algebra of*

constructions (SEQ, PROD, recursive substitution) determines the *algebra of generating functions*, and the exponent comes from a single eigenvalue derivative. For series-parallel Φ the transformation is degenerate and $\lambda^\sharp = s$ (hence $d_f = 1$); for non-series-parallel Φ the eigenvalue carries non-trivial graph information. The obstruction at $\mathbb{Z}^d$ is precisely that there is no finite rewrite Φ that generates $\mathbb{Z}^d$ by self-similar substitution: translation invariance is a *limit* construction rather than a compositional one, so the symbolic-method machinery has no finite animal to seed.

Two further animal experiments and their negative findings. Having exhausted the self-similar hierarchical approach, the natural next question is whether a tower of *non-self-similar* but still algebraic animals can close in on $d_f^{(3)}$. Two are implemented and documented here for completeness; both fail, and the failure mode of each is informative.

(a) *Finite-cross-section tubes* (`rdft/ac/lerw_tube.py`). LERW on the tube $\{0, \dots, N-1\}_{\text{periodic}} \times \{0, \dots, L-1\}_{\text{Dirichlet}}$ defines, for each N , an N^2 -state Markov chain with an algebraic spectrum, hence an algebraic candidate exponent $d_f^{(N)}$. Monte-Carlo extraction at $N = 1, 2, 3, 4, 5$ with $L \in \{8, 12, 16, 24, 32\}$ gives $d_f^{(N)} \approx 1$ for every N tested (ballistic collapse). The reason is structural: a narrow tube is quasi-1D, and LERW on a 1D path is a deterministic monotone traversal. Cross-sectional wandering contributes a sub-leading constant factor, not a different power of L . The tower of tube exponents does not approach $d_f^{(3)}$ as N grows — it stays at 1 until $N \sim L$, at which point the “tube” is no longer a tube and one is simply looking at an isotropic box. This rules out the tube-transfer-matrix approach as a tower of algebraic animals for $d_f^{(3)}$.

(b) *Algebraic extrapolation on isotropic boxes* (`rdft/ac/lerw_extrap.py`). The correct target observable is the isotropic-box mean length $\langle |\gamma| \rangle_L$ on $\{0, \dots, L-1\}^3$, which our Dirichlet sampler produces. Classical AC-style algebraic extrapolation applied to the sequence $\{\langle |\gamma| \rangle_L\}$ includes:

- Naive log-log fit;
- $(1 + B/L^\omega)$ correction-to-scaling fits with ω fixed or grid-searched;
- Richardson extrapolation on effective exponents $d_f^{\text{eff}}(L) := \log(\langle |\gamma| \rangle_{L+} / \langle |\gamma| \rangle_{L-}) / \log(L_+ / L_-)$, assuming a $1/L$ leading correction;
- Neville–Richardson tableau extrapolation, assuming an integer-power correction tower $1/L, 1/L^2, \dots$

On clean synthetic data with a known correction structure the schemes recover the injected exponent (tests witness this). On real LERW data in $d = 2$ and $d = 3$, however, the “sophisticated” schemes do *not* outperform the naive fit: they systematically bias d_f downward because the true correction to scaling is not a simple power of $1/L$. In $d = 3$ at $L \in \{4, 6, \dots, 16\}$ the naive fit gives $d_f \approx 1.62$ (coincidentally within

a few parts per thousand of Wilson’s 1.6236), while the $(1 + B/L)$ correction fit drags it to ≈ 1.54 and Richardson extrapolation to ≈ 1.29 . This is a concrete epistemic statement: *LERW finite-size corrections do not live in the algebraic-power-of- $1/L$ category that Richardson and Neville assume*. If they did, those schemes would dominate the naive fit by orders of magnitude. They don’t, so the corrections include some non-algebraic structure (logarithmic factors, non-integer powers, or boundary-specific terms that do not obey the $\Phi^\sharp$ -eigenvalue story).

A fifth animal: the sandpile group (`rdft/ac/sandpile_group.py`). The Majumdar–Dhar bijection (Majumdar and Dhar, 1992) maps recurrent configurations of the abelian sandpile model (ASM) on a graph G to spanning trees of G ; Wilson’s theorem then maps the a -to-sink path in the UST to LERW. So ASM states, spanning trees, and LERW carry the same probabilistic content. The algebraic upgrade is that recurrent ASM configurations form a finite abelian group — the *sandpile group*

$$K(G) = \mathbb{Z}^{V \setminus \{s\}} / \Delta' \mathbb{Z}^{V \setminus \{s\}},$$

with Δ' the Laplacian with the sink row/column removed. The sequence $\{K(\mathbb{Z}_L^d)\}_{L=1,2,\dots}$ is a concrete tower of finite algebraic animals, each of which is completely determined by the Smith normal form of Δ' .

Computation (via integer Smith normal form): for $\mathbb{Z}_L^2$ the non-trivial invariant factors are $L = 3$: $[6, 6, 18, 18]$; $L = 4$: $[2, 2, 8, 24, 24, 24, 96]$; $L = 5$: $[10, 10, 50, 50, 50, 50, 50, 50]$; for $\mathbb{Z}_3^3$ torus, $[6, 6, 6, 18, 18, 54, 54, 54, 54, 162, 162, 162]$. The group orders (= Matrix-Tree spanning-tree counts) are verified against the closed-form Laplacian-eigenvalue product, so the computation is correct.

Fifth failure mode. The group structure carries clear number-theoretic content (the divisors are products of prime powers determined by L ’s factorisation), but its scaling is erratic rather than universal: the largest invariant factor grows as $18, 96, 50, \dots$ for $L = 3, 4, 5$, giving effective exponents $\log(d_{\max})/\log L \in \{2.63, 3.29, 2.43\}$ — dominated by number-theoretic fluctuations rather than any clean power-law. The obstruction is structural: the sandpile GROUP is the Smith normal form of Δ' , a *static* algebraic invariant. The LERW *fractal dimension* is a dynamical observable — the expected length of the UST path between two points — that depends on how the full Laplacian spectrum interacts with the path decomposition, not on the Smith-normal-form decomposition alone. What WOULD carry d_f is the avalanche-size / wave-profile distribution of the ASM *dynamics*; that is not an algebraic invariant of $K(G)$.

The upshot: the Majumdar–Dhar bijection is real and deep, and the sandpile group is a legitimate algebraic animal for every finite graph, but it is the wrong algebraic object to extract $d_f^{(3)}$ from. The group’s content is Kirchhoff (spanning-tree count, hence bulk entropy); it does not carry path-length scaling. This is another honest no-go to add to the catalogue.

What IS the right combinatorial object for path length? (`rdft/ac/lerw_exact.py`) With five algebraic animals catalogued as failures, a sharper question is: *what is the combinatorial object that actually captures dynamical path length, the thing we ought to have been extracting asymptotics from?* The answer has been in the paper from the start: it is the length generating function of Prop 1 itself,

$$F_{ab}(z; G) = \sum_{\gamma: a \rightarrow b \text{ simple}} z^{|\gamma|} \frac{w(\gamma) \tau(G/\gamma)}{\tau(G)}. \quad (19)$$

$F_{ab}(z; G)$ is a *rational function* in z on every finite graph G , with mean length $\langle |\gamma| \rangle = F'_{ab}(1)$. This is deterministic — no Monte Carlo — and is the exact algebraic object whose asymptotic behaviour in L would give $d_f^{(d)}$.

Two sanity checks. Naive hope: maybe $\langle |\gamma| \rangle$ equals the sum of absolute electric currents $\sum_e |i_e(a \rightarrow b)|$, where i_e is the unit current on edge e in the resistor network with unit edge conductances. This is true on trees. On K_4 from vertex 0 to 1 the sum of absolute currents is $\frac{1}{2} + 4 \cdot \frac{1}{4} + 0 = \frac{3}{2}$ while the exact mean from Eq. (19) is $13/8 = 1.625$ — the discrepancy comes from edge (2, 3), which carries zero current by symmetry but still appears in $1/8$ of LERW paths. So the correct combinatorial object is *not* a sum of currents; it is the Kirchhoff ratio summed against $z^{|\gamma|}$ over simple paths, exactly as in Eq. (19).

Deterministic values from Eq. (19) via direct path enumeration plus Kirchhoff determinants:

Case (corner-to-corner)	Exact $\langle \gamma \rangle$	MC check
$\mathbb{Z}_3^2$	$17/4 = 4.25$	4.26 ± 0.01
$\mathbb{Z}_4^2$	$6.69228\dots$	6.69 ± 0.02
$\mathbb{Z}_5^2$	$9.29526\dots$	9.30 ± 0.02
$\mathbb{Z}_2^3$	$53/16 = 3.3125$	3.31 ± 0.02
$\mathbb{Z}_3^3$	$7.84146\dots$ (~ 2 min enum.)	7.85 ± 0.05

These are *exact rationals* (the last two decimalised for brevity), and they match Monte Carlo to sampling precision. The length generating function is therefore correctly identified as the combinatorial object for path length.

Why we didn’t already extract $d_f^{(3)}$ from it. Two obstructions — the same two that shaped the whole paper:

- **Computation cost.** The number of simple paths on $\mathbb{Z}_L^3$ grows super-exponentially in L ; direct enumeration becomes infeasible beyond $L = 3$. No known analytic shortcut reduces the Kirchhoff-ratio sum to anything D-finite as $L \rightarrow \infty$ — because the scaling limit IS the non-D-finite lattice Green’s function.
- **Structural obstruction.** Substituting $z = 1$ in Eq. (19) gives 1 (normalisation); the derivative at 1 is a *weighted resistance-like quantity* against the full Laplacian spectrum, and extracting its L -scaling is exactly the closed-form problem we cannot solve in $d = 3$.

So the right combinatorial object is in our hand; the rate-limiting step is not identifying it but extracting its asymptotics at scale. The five algebraic animals catalogued above are the known attempts at short-cut asymptotics, each losing a different ingredient of $F_{ab}(z; \mathbb{Z}_L^d)$ in the process. A wider AC category (resurgent transseries, non-D-finite closure, determinantal / Fredholm approximation of the scaling limit) is what would be required to close the extraction — and that is the honest open direction this paper hands off.

Notebook: a special deep problem. The two negative results above, taken with the MK diamond, Sierpiński gasket, and sandpile-group animals, catalogue five attempted algebraic animals for $d_f^{(3)}$:

Animal	Failure mode
MK diamond (4, 2)	Correct $d_{\text{eff}} = 3$ but series-parallel; UST is trivially ballistic. $d_f = 1$.
Sierpiński gasket	Correct loop-erasure structure but wrong fractal dimension ($d_H = \log_2 3$, not 3). $d_f \approx 1.20$.
$\mathbb{Z}^3$ tube tower	Quasi-1D at every finite N . $d_f^{(N)} = 1$ for all accessible N .
Algebraic extrapolation on Dirichlet boxes	Naive fit hits ~ 1.62 , but no scheme improves on it — corrections are not a clean algebraic power of $1/L$.
Sandpile-group tower (Majumdar–Dhar bijection)	Right bijection to LERW; group carries only Matrix-Tree content (spanning-tree count), not path-length scaling. Erratic $\log d_{\text{max}} / \log L$.

The five failures have a common theme: they each succeed at capturing *one* of the ingredients of $\mathbb{Z}^3$ (dimension, loops, finite Markov structure, algebraic corrections, or the Majumdar–Dhar algebraic-group upgrade) while losing the dynamical path-length content that d_f actually measures. The target object sits at the intersection of all five, and no finite algebraic construction we have been able to put to it captures the intersection.

This is a concrete statement of the “deep” part of the problem — stronger than “we don’t have a closed form”. It says: the generating-function category within which we can do AC asymptotic extraction is populated by objects, none of which contains the $\mathbb{Z}^3$ LERW two-point function in its closure under composition. Whether a *wider* AC category (resurgent transseries, hyperasymptotic Borel summation, non-D-finite closure) contains it is the honest open question this paper leaves open.

What the experiments do and do not demonstrate. Each module now has a concrete computational handle tested against an independent reference:

- LERW: samplers consistent with the exact Kirchhoff ratio, and numerical recovery of Kenyon’s $d_f = 5/4$ in $d = 2$ and Kozma’s $d_f \approx 1.624$ in $d = 3$ from a Dirichlet-box Lawler sweep (each within $\sim 3\%$ at modest sample size); plus the combinatorial-animal experiment above, which quantifies exactly what self-similar surrogates capture and miss about $\mathbb{Z}^3$.

- KPZ: a transfer-matrix replica partition function consistent with the CRN enumeration and with the free-walker spectrum; the exact $\lambda(n, \beta; W)$ curve at $n = 1, 2, 3$ shows the Bethe-ansatz cubic-in- n shape, with $D(3)/D(2) \rightarrow 4$ (Kardar continuum limit) monotonically as W grows; and the closed-form 2-body replica rate $D(2, \beta; \mathbb{Z}) = 2\beta^2 - \log(2e^{\beta^2} - 1)$ (Thm. 6) is the Tier-A deliverable for the $n = 2$ sector.
- TAP: the Catalan-series reconstruction of the semi-circle resolvent works to the precision of the truncation.

The open exponents and coefficients are still open. What these experiments show is that the CFAC enumeration layer produces concrete, testable objects in every case — not just structural identities.

8 What the enumeration layer buys

Prop. 1, 3, and 4 are not new mathematics. The point is that they share a common CFAC-enumerative structure:

Case	Enumerated object
LERW 3D	Simple paths weighted by $\tau(G/\gamma)/\tau(G)$
TAP complex-ity	Rooted one-face maps by genus (Harer–Zagier)
KPZ moments	n -tuples of walkers by coincidence count

Each is a weighted sum over a discrete combinatorial set, with weights that are either first Symanzik polynomials (Eq. (1)), ribbon-graph Euler numbers (Eq. (9)), or Doi–Peliti vertex factors (Eq. (3)). The existing CFAC pipeline handles all three.

What the enumeration layer does *not* buy.

- For LERW, the 3D exponent ν_3 is still a hard scaling limit of the finite-graph formula; no closed form falls out.
- For TAP, the genus expansion organises finite- N corrections but does not by itself give the complexity density Φ_0 in closed form for general mixed p -spin models.
- For KPZ, integer- n replica enumeration is a classical weighted walker count; the open problem (upper-tail coefficient universality) lives at the $n \rightarrow 0$ continuation, outside the enumeration.

What it does buy is a concrete combinatorial definition of the object under study, with finite-size exactness, and a clean statement of where the stratification theorem would have to be extended (scalar $\rightarrow$ matrix bridge; algebraic $\rightarrow$ non- D -finite GF; deterministic $\rightarrow$ disorder-averaged vertex).

Where this leaves the scope map. Combining this note with (Amarteifio, 2026b, §7, §8) gives a two-tier scope statement:

Tier A (stratification theorem): polynomial DSEs with algebraic / D -finite generating function and a scalar bridge. Delivers a skeleton exponent $\tau_0 = 1 + 1/k$ plus a computable one-loop dressing $\gamma_k(d)$.
Tier B (enumeration): any weighted combinatorial object (Feynman diagrams, spanning trees, ribbon graphs, n -walker paths, partition sums) with Symanzik / Laplacian / Schur structure. Delivers exact finite-size generating functions and explicit rational / determinantal representations.

The three boundary cases of (Amarteifio, 2026b) fall outside Tier A but strictly inside Tier B. The scope-claim “CRN-in-field is the underlying object” in (Amarteifio, 2026b, §8) is honest at Tier B; the stronger claim that the stratification theorem applies is honest only within Tier A.

9 Discussion

The three propositions above are classical results rewritten in CFAC enumeration language. The value of the rewrite is that it makes the obstruction for each case *localisable*: LERW 3D needs non- D -finite resummation of a Kirchhoff ratio; TAP complexity needs a matrix-valued extension of the scalar bridge to close the complexity density into closed form for general p -spin; KPZ upper tails need a disorder-averaged bridge to fix the coefficient from the Schur edge.

Two further cases that the companion paper lists only in passing — BRST-quantised gauge theory and replica $n \rightarrow 0$ — fit the same template. BRST cohomology is a graded enumeration of ghost-number sectors; replica $n \rightarrow 0$ is a continuation of an enumeration. Neither is a Tier-A object. Both are, at the enumeration level, concrete combinatorial constructions the pipeline can handle at fixed n or fixed ghost number.

Next steps. The three modules of §6 establish the baseline; the interesting extensions are: (i) push the LERW Kirchhoff-ratio computation to $\mathbb{Z}_N^3$ for $N \lesssim 20$ via a Fourier-diagonalised Laplacian, and track the finite-size scaling of $\partial_z \log F_{0b}$ against the Kozma exponent; (ii) extend the genus-graded ribbon-graph enumeration to GOE (non-orientable surfaces, Goulden–Jackson coefficients) and plug it into the Kac–Rice Legendre transform to recover the p -spin complexity density as a map-count resummation; (iii) implement the Schur-process reduction Eq. (13) for integrable polymer geometries and verify it against the direct CRN enumeration on small instances (e.g. $N \leq 4$, $n \leq 3$).

None of these closes the open exponent or coefficient. Each sharpens the line between what CFAC-as-theorem covers and what CFAC-as-enumeration covers, and supplies a concrete object for downstream work.

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